

# Embodying Relationships: Designing TUIs for Co-Located Human-Human Dynamics

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## Abstract

This hands-on workshop invites participants to explore how Tangible User Interfaces can be designed to enrich meaningful co-located human-human dynamics. We invite participants to build on prior work with TUIs for task completion and knowledge sharing, and to consider how tangible and embodied qualities might also shape relationships and the ways people relate to one another. Human social interactions are characterized by unique dynamics, whether rooted in trust, tension, reciprocity, perspective-taking, empathy, or other subtle concepts that are often difficult to articulate and even harder to enact. When carefully coupling psychological theory with TUI design principles, TUIs, through their spatial and embodied qualities, have the potential to make these processes more concrete, enhancing human-human communication and relationships. In this workshop, participants will create low-fidelity prototypes by sketching, folding, cutting, and assembling artifacts that embody psychological and relational dynamics. The workshop aims to generate novel ideas, stimulate critical discussion, and open new directions for how technology can be integrated into physical spaces in subtle, embodied ways that enhance meaningful human-human dynamics.

## CCS Concepts

• **Human-centered computing** → *Haptic devices*; Heuristic evaluations.

## Keywords

Tangible user interface, Co-located interaction, Human-human dynamics

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## 1 Motivation Background

This workshop explores how Tangible User Interfaces (TUIs) can be designed to enrich meaningful human-human dynamics and connection in co-located interaction. We define meaningful dynamics as interactions in which two individuals strive to strengthen their bond, marked by the ability to talk explicitly about both the communication itself and the relationship [18]. Such dynamics can be found not only in romantic partnerships, but also in close friendships, collegial bonds, and family ties such as parent-child or sibling relationships.

In HCI, and particularly in research on TUIs, social interaction is often described in terms of outcomes within a given context, typically toward a shared goal such as collaboration, task completion, or learning [13]. Most TUI work centers on education (e.g. [8]) and collaboration (e.g. [12]), where the human-human relationship itself is not the primary focus, and interaction is evaluated

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mainly through performance or engagement rather than relational dynamics.

In this workshop, we extend beyond this view by drawing on relational communication theory, which emphasizes how verbal and nonverbal messages define the relationship itself [5]. From this perspective, TUIs offer opportunities to surface and shape the relational qualities of interaction—such as trust, reciprocity, perspective-taking, and intimacy—rather than focusing solely on observable outcomes.

From the earliest stages of human culture, people have relied on physical objects and embodied action to engage with abstract ideas. The abacus, for instance, transformed numbers into beads and gestures that could be directly manipulated. TUIs build on this tradition, grounded in the idea that physical interaction enhances cognition [3]. By embedding shapes, materials, and spatial configurations into interfaces, TUIs can support comprehension and meaning-making through embodied engagement. Leveraging physicality both as a manipulable interface and as a mode of representation [22, 23], TUIs have the potential to translate abstract processes into concrete experiences that strengthen understanding [1].

This becomes especially powerful when TUIs are designed around body-based schemas, drawing on embodied experience as the foundation for interaction [2]. By grounding thought in physical form, TUIs create a bridge between cognition and action, making abstract processes more accessible through tangible engagement [6, 14]. Concepts such as flow, hierarchy, or rhythm can, for example, be represented through weight, motion, or spatial arrangement [4, 28].

Building on their embodied and representational qualities, TUIs have been explored across diverse domains. In education, they promote experiential learning and deepen engagement [8]. In assistive technology, they provide alternative or complementary channels to verbal communication [15, 25]. TUIs have also been integrated in collaborative settings. Their physicality, which provides multiple entry points, can strengthen interaction and shared awareness that support mutual understanding [12, 13, 26]. In recent years, researchers have also examined how technology in general [24] and specifically tangibles can support intimacy, affect, emotions, and moods.

This workshop builds on this emerging body of work and positions TUIs as a lens for rethinking how technology can support everyday co-located human–human dynamics, such as those between partners, family members, friends, or colleagues. One such example is the Awareness Object, a tangible artifact designed to enhance awareness and reflection in parent–child collaboration, created around the mentor–peer scale that represents different parental roles during joint activities [20]. Its visible, tangible presence lowered the threshold for communication between parents and children about the roles parents adopt in these interactions (see Figure 1 on the left) [20]. At the same time, we also look beyond TUIs to physical and embodied interventions more broadly, showing how it can mediate human–human dynamics in diverse ways. For instance, robotic bar stools rotating couples towards one another during a conflict facilitated a change in body orientation and eye contact between the couples. Such changes in orientation and gaze, were interpreted as non-verbal cues that encouraged intimacy, reduced the intensity of the conflict, and facilitated hugs and kisses (see Figure 1 on the middle) [21]. Similarly, placing a

non-humanoid robot between couples during conflict discussions increased their awareness of the conversation and its tone. By displaying gestures of curiosity when calm and fear when heated, the object subtly encouraged emotional regulation without disrupting communication (see Figure 1 on the right) [11].

These studies show that by externalizing complex emotional and social dynamics and providing a shared reference point, physical and embodied interventions, including TUIs, can lower cognitive effort, anchor individuals in shared experiences, and open additional non-verbal channels for communication. Through guided discussion and rapid prototyping, this workshop will invite participants to experiment with embodying relational concepts in tangible and physical forms, contributing to new design directions for enriching co-located human–human dynamics.

## 2 Organizers

The organizing team brings together diverse expertise in HCI, psychology, design, and fabrication, combining theoretical insight with hands-on practice. Their complementary backgrounds ensure a workshop that is both conceptually rigorous and applicable to design outcomes, offering participants rich perspectives on tangible and embodied interaction.

**Ofir Sadka** is a PhD candidate at the Faculty of Industrial Engineering and Management at the Technion, in collaboration with the Media Innovation Lab at Reichman University. With a background in Psychology and HCI, his research explores the intersection of physical embodied interaction and meaningful human–human communication. He focuses on how tangible devices can make abstract relational processes concrete, creating an additional non-verbal communication channel that complements, replaces, or emphasizes verbal communication.

**Iddo Wald** is a research fellow at the Digital Media Lab, University of Bremen. His research lies at the intersection of HCI, neuroscience, and embodied interaction, with a focus on sensory augmentation and tangible interfaces. With a background in interactive systems and design, he explores how materiality, embodiment, and sensory technologies can shape human experience, creativity, and social connection.

**Andrey Grishko** is the Design and Fabrication Lead at the Media Innovation Lab (milab) at Reichman University. Trained as an industrial designer, he leads the ideation, design, and prototyping processes that bring research concepts into physical form. His expertise spans traditional carpentry, digital fabrication, 3D printing, and electronics, enabling the creation of interactive artifacts that bridge design, technology, and research. At the milab, he collaborates closely with researchers and students to translate theoretical ideas into tangible prototypes that support experimentation, user studies, and public exhibitions.

**Ken Nakagaki** is an Assistant Professor at the University of Chicago, where he directs the Actuated Experience Lab (AxLab). His research explores the future of user experience through actuated user interface technologies that enable shape-change and physical reconfiguration. With a background in physical computing, robotics, and HCI, he investigates how computation can be coupled with the physical environment and materials to create novel tangible and



**Figure 1: Left: The Awareness object during collaborative activity between parent-child couple, lowered the threshold for communication about the roles parents adopt in these interactions (used with permission [20]). Middle: A couple during conflict after inward movement of the robotic bar stools were interpreted as non-verbal cues that encouraged intimacy. (used with permission [21]). Right: A non-humanoid robotic object between couples during conflict discussions increased their awareness of the conversation and its tone (used with permission [11]).**

embodied experiences. His work speculates on and invents at the intersection between our fingertips and the physical world.

**Julia Dominiak** is a PhD student jointly affiliated with Lodz University of Technology and Technische Universität Wien, with a background in computer science and control engineering. Her research focuses on tangible interaction and digital fabrication, and she has strong technical experience in designing and engineering interactive systems, including a robotic system developed to support arachnophobia exposure therapy.

**Tanja Döring** is a Full Professor at the Institute of Psychology and Ergonomics at Technische Universität Berlin, where she leads the Interactive Human-Technology Systems lab. Her research explores HCI with a particular focus on materiality, tangible and embodied interaction, and AR/VR technologies. She applies these approaches across diverse domains, including learning, health, creativity, and social interaction. Driven by a vision of creating technologies that foster human potential and connection, she seeks to design interactive systems that empower people and enhance their experiences.

**Oren Zuckerman** is an HCI researcher and designer at Reichman University's School of Communications. Oren research how people interact with physical-digital technologies, specifically social robots, tangible interfaces, and smart materials. Oren is the academic director of the MA in HCI, the GMBA in Innovation and Entrepreneurship, and the milab research and prototyping lab. Oren's interest in abstract movement and design has been inspired by the works of Alexander Calder, Bruno Munari, the Bauhaus movement, and Pina Bausch.

### 3 Workshop Activities and Length

The workshop is designed as a 90-minute interactive session that includes discussion and hands-on prototyping:

- Introduction of workshop goals, followed by a brief presentation of the human-human interaction scenarios described in participants' papers. We will introduce core TUI principles: affordances [10, 16], constraints [9, 17], distributed cognition

[19], embodiment [6], visibility and locality [27], and peripheral interaction [7] to illustrate how material and embodied properties can guide action and support human-human dynamics without overshadowing them.

- Choosing a relational dynamic, theory and context: Participants will select a psychological strategy and relational dynamic relevant to the human-human interaction they wish to focus on. To support this process, the organizing team will prepare a set of strategies, definitions, and academic references in advance, drawing from the papers submitted by participants.
- Participants then select a low-fidelity tangible prototype, a minimal physical form that acts as a thinking prompt for imagining dyadic TUI interactions. For instance, balance may embody mutual support or negotiation and weight may represent responsibility or emotional load. Using quick, accessible methods such as sketching, folding, cutting, and assembling simple materials such as paper, cardboard, pins, and glue, participants will modify low-fidelity prototypes that externalize these strategies. The emphasis will be on speed and creativity rather than refinement, enabling concepts to be rapidly materialized, shared, and discussed. Participants who prefer not to work with physical materials may also sketch or storyboard their ideas, ensuring accessibility and inclusivity.
- Closing Discussion: Participants will reconvene to share their prototypes and reflections. Together, we will map out a potential design space for supporting co-located human-human dynamics with TUIs. The session will also reflect on the ethical and methodological challenges of evaluating such designs—acknowledging the sensitivities involved in studying relationships and the complexities of capturing relational dynamics beyond surface-level interaction. The discussion will conclude by identifying future directions for research and design, and by considering how these early explorations can inform the development of technologies that enrich co-located relationships.

## 4 Post Workshop Plans

Drawing on the prototypes, design outcomes, and reflections produced during the workshop, the post-workshop efforts will support the workshop's two goals: (1) cultivating an active community around this emerging space and (2) establishing a clear research agenda for TUIs that support human–human dynamics focusing on two elements. To advance the second goal, we will focus on two directions:

- Translation of single-User TUI principles to dyadic contexts: We will examine how established TUI principles, such as embodied interaction, affordances, and manipulation manifest when applied to dyads rather than individuals. The paper will identify where these principles carry over, where they shift, and where new principles or adaptations are needed to address relational dynamics between two people.
- Synthesis of opportunities, challenges, and open questions: Based on workshop discussions and artifact analysis, we will articulate the key opportunities and tensions that arise when designing TUIs for dyads. This includes emerging themes, recurring interaction patterns, breakdown points, and promising directions for future research. This synthesis will form the foundation for a broader research agenda in TUI-mediated human–human interaction.

Together, these outcomes will position the workshop as a catalyst for a growing body of TEI research that moves TUIs beyond collaboration and learning toward a richer understanding of how they can shape, scaffold, and enhance interpersonal dynamics.

## 5 Accessibility

To ensure accessibility and inclusivity, participants who prefer not to work with physical materials will also be able to sketch, storyboard, or prototype concepts using online collaboration platforms. The workshop itself will be held as an in-person session at CHI 2026. The physical space and facilities will follow the accessibility guidelines established by the CHI Accessibility Chairs, ensuring that all participants can engage fully in the activities. Organizers will remain attentive to participants' individual needs and will provide alternative ways of contributing to discussions and prototyping when necessary.

## 6 Call for Participation

This workshop explores how Tangible User Interfaces (TUIs) can mediate meaningful co-located human–human dynamics, enriching relationships between people situated in the same space. By dynamics we refer to the evolving relational qualities between people, such as trust, reciprocity, perspective-taking, or intimacy, that shape and sustain their relationships. We invite participants to build on prior work with TUIs for task completion and knowledge sharing, and to consider how tangible and embodied qualities might also shape relationships and the ways people relate to one another.

The workshop will combine discussion with hands-on activities. Participants will identify human-human contexts and choose relevant strategies or relational qualities. They will rapidly prototype artifacts using low-fidelity methods like sketching, folding, and simple craft materials. No prior experience required. These prototypes will serve as prompts for sharing reflections on how TUIs

might intervene in relational processes. We welcome researchers, designers, and practitioners interested in embodied interaction, tangible design, and human communication. Our target participants include members of the TEI community, as well as researchers and practitioners with psychology or relationship-science backgrounds who are curious about how tangible technologies can enhance or shape co-located human–human dynamics.

To participate, please submit a short position paper (1–2 pages in ACM format) describing your background and a human–human interaction you wish to explore through TUIs. Submissions will be reviewed for relevance and diversity of perspectives. Accepted papers will be published in open access on the workshop website. At least one author of each accepted paper must register for and attend the workshop. We expect around 20–30 participants. For more details: <https://andgris.wixsite.com/embodying>.

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